Temujin Orkhon

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EDUCATION

Massachusetts Institute of Technology — **Expected June 2026**
B.S, Double Major in Chemistry & Biology and Physics, Minor in Computer Science — Cambridge, MA
- GPA: 5.0/5.0
- Relevant Subjects: Computational Chemistry; Fundamentals of Chemical Biology; Machine Learning; Programming in Python; Quantum mechanics

RESEARCH EXPERIENCE

Learning Matter Lab (Gómez-Bombarelli Group) — **January 2023 – June 2024**
Undergraduate Researcher — Cambridge, MA
- Constructed multi-layered machine learning models, utilizing uncertainty as a learnable parameter, achieving greater generalizability and accuracy in predicting molecular property
- Developed software aimed at generating artificial datasets, implementing, and benchmarking existing molecular property machine learning models

Elkin Lab — **January 2025 – June 2025**
Undergraduate Researcher — Cambridge, MA
- Designed novel algorithms to extract organic transformation templates, improving on literature template extractors through the usage of context dependent reaction centers
- Helped in the development of software that streamlines the analysis of total synthesis data to determine impactful multistep transformations for further experimental optimization

Van Voorhis Group — **June 2024 – Present**
Undergraduate Researcher — Cambridge, MA
- Developing electronic structure methods employing quantum embedding to calculate the electronic properties of strongly correlated chemical systems.
- Simulating and analyzing defected semiconductor materials for applications in quantum computing, collaborating with outside universities

PUBLICATIONS & PRESENTATIONS

- **Orkhon T.**; Weatherly S.; Van Voorhis T. (2025), "Extracting band structures using Bootstrap Embedding". Article in preparation.
- **Contributor** (2025), "QuEmb: A Toolbox for Bootstrap Embedding Calculations of Molecular and Periodic Systems", Journal of Physical Chemistry, doi.org/10.1021/acs.jpca.5c02983
  Contributed by testing the periodic bootstrap embedding code, and helping with bug fixes
- **Orkhon T.**; Rein J.; Elkin M. (2025), "Holistic reaction template generation using a context-driven approach" Presented at the Chem UROP Symposium, Cambridge, MA
- **Orkhon T.**; Greenman K.; Green W.; Gómez-Bombarelli R. (2024) "Uncertainty improves multi-fidelity machine learning", Presented at the ChemE UROP Symposium, Cambridge, MA
- Greenman K.; **Orkhon T.**; Green W.; Gómez-Bombarelli R. (2023), "Multi-fidelity deep learning for data-efficient molecular property models from experimental and computational data", AIChE Annual Meeting

TEACHING EXPERIENCE

MIT Department of Chemistry — **February 2024 – Present**
Chemistry Tutor — Cambridge, MA

- Provided one-on-one tutoring in general chemistry, organic chemistry, and physical chemistry to MIT students interested in chemistry.
- Created custom lesson plans, practice problems, and review materials tailored to the student's needs.

Talented Scholars Resource Room **September 2023 – June 2024**
Facilitator Cambridge, MA

- Facilitated student groups, providing comprehensive assistance with problem sets and hosting exam reviews, improving academic performance of students at MIT.
- Delivered personalized one-on-one guidance to MIT students in fundamental introductory subjects, ensuring an effective approach.

TomYo EdTech **August 2023 – September 2024**
College Counselor Ulaanbaatar, Mongolia

- Advocated to Mongolian students to study abroad, guiding high school students through the college application process.
- Counseled prospective students on their higher education plans, offering valuable insights and guidance to align their goals with suitable academic pathways.

HONORS & AWARDS

- Best poster presentation in the ChemE UROP Symposium (2024)
- International Chemistry Olympiad Silver medalist (2021, 2022)
- International Mendeleev Chemistry Olympiad Silver medalist (2021)

TECHNICAL & LABORATORY SKILLS

- **Computational**: Python, Pandas, Numpy, Pytorch, SciPy, FHI-aims, PySCF, QuEmb, QChem
- **Laboratory**: Titration, Distillation, Thin-Layer Chromatography, Column Chromatography, UV-vis Spectroscopy, Fluorescence Spectroscopy, FTIR Spectroscopy